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When trust is lost: the impact of interpersonal trauma on social interactions

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Abstract

Background. Trauma due to deliberate harm by others is known to increase the likelihood of developing Post-Traumatic Stress Disorder (PTSD). This is the first study investigating basic and dynamic trust in 'interpersonal' PTSD.

Methods. Thirty-two participants with PTSD and 22 healthy controls played a novel multi-round version of a monetary investment protocol, the so-called 'Trust Game', a task from the behavioural economics literature, which is considered to involve trust and reciprocity. We used two 'Trust Games' including cooperative and unfair partners.

Results. Findings showed an effect for lower basic investment in PTSD compared to healthy controls, that trended towards significance ($p = 0.09$). All participants showed behavioural flexibility and modified their trust based on behavioural cues from their cooperative and unfair game partners. However, participants with PTSD made significantly lower investments towards the cooperative partner than controls. Investments towards the unfair partner did not differ between groups. Higher trauma scores were associated with lower levels of trust-related investments towards the cooperative but not the unfair game partner.

Conclusion. The association between reduced trust towards cooperative others in individuals who experienced interpersonal trauma could indicate acquired insensitivity to social rewards or inflexible negative beliefs about others as a sequel of the traumatic experience, which increases in a dose response relationship with the severity of the trauma. A specific focus on cooperation and trusting behaviour could provide a treatment target for future cognitive and pharmacological interventions.

Introduction

Traumatic events can take many forms, such as experiencing or witnessing sexual or physical assault, a terrorist attack, war, or a natural disaster. Most recover, but a proportion will develop Post-Traumatic Stress Disorder (PTSD), leading to a lifetime prevalence of about 7% (Kessler *et al.*, 2005). Individuals whose trauma was 'interpersonal' (caused deliberately by other people) are twice as likely to develop PTSD, relative to those who experience accidental trauma with more complex and severe symptoms (Charuvastra and Cloitre, 2008).

Theoretical models of trauma suggest that an individual's interpretation of others' intentions is more important than the actual responses themselves (Ehlers and Clark, 2000). Not surprisingly, interpersonal trauma can alter a sense of trust in a way that is likely to be a significant contributor to relationship difficulties. Cias *et al.* (2000) reported the only distinguishing variable from PTSD and a subsample of individuals with depression were lower levels of interpersonal trust. Trust is widely considered fundamental for the recovery of trauma survivors by enabling them to effectively manage conflict in relationships and establish mutually cooperative interactions (e.g. Williams *et al.*, 2014; Chouliara *et al.*, 2017). Trust can be defined as '*a psychological state comprising the intention to* accept *vulnerability based upon the positive expectations of the intentions or behaviour of* another' (Rousseau *et al.*, 1998, p. 395). The vulnerability may be defined by the costs and benefits of trusting (i.e. costs are the losses incurred from betrayal *v.* profits when the trustee reciprocates). The impairment in trust can take different forms that may depend on the nature of trauma. Those who are exposed to interpersonal trauma are more likely to report a generalising of mistrust to all people, irrespective of whether these individuals had done anything to warrant this suspicion; they are also more likely to distance themselves from others, not giving others a chance to prove trustworthiness. Clinical experience indicates PTSD patients typically wish they could trust others more, and often describe themselves as having been too trusting before their experience of traumas, and thus view mistrust as a necessity to their continued sense of safety and control. Mistrust can form quickly, with minor breaches of trust being evidence confirming an implicit

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suspicion of others. Such thoughts may vary across a continuum, from low levels of mistrust and suspiciousness to distressing unfounded paranoid ideas (Freeman *et al.*, 2010). A less trusting attitude towards others may be initially self-protective, yet longer-term the reduced ability to establish trust can be a barrier to engagement with mental health services.

Quantifying relational trust in an objective way is not an easy task since reasons to trust or not will vary between individuals. One objective approach to quantify trust can be repurposed from economic game theoretical approaches (Camerer, 2003), without bias created by the inherent contextual factors in questionnaire based studies. The interactive 'Trust Game' (Berg *et al.*, 1995; King-Casas *et al.*, 2005) attempts to assess dynamic changes in the development of trust-related behaviour during the investment and reciprocation of money between an investor and partner; the cross-diagnostic value of this paradigm in the investigation of social behaviour and its associated neural mechanisms was recently reviewed by Fett *et al.* (2015). The participant has to decide how much of an endowment s/he wants to invest, and this transferred amount is multiplied and the partner decides how much of the total amount to return. There are two game partners: cooperative and unfair, which allows exploration of whether the participant can reciprocate investment behaviour, a possible index of trust, following repeated friendly signals by the partner, or whether there is a general insensitivity to social feedback. The 'Trust Game' has previously been used in other clinical (King-Casas *et al.*, 2008; Fett *et al.*, 2012*a*, 2012*b*; Zhang *et al.*, 2012; Fett *et al.*, 2014) and non-clinical populations (Houser *et al.*, 2010). To date, there is surprisingly little research on the link between trust and PTSD and only a few studies have adopted the 'Trust Game' with PTSD samples. For example, Cisler *et al.* (2015) found women with PTSD who had been assaulted responded to an unfair playing style with decreases in trust and showed significantly lower learning rates suggesting less flexibility in updating social expectations and less use of social experiences in guiding social decision-making. Within a therapeutic context, Williams *et al.* (2014) showed changes in interpersonal trust with 21 veterans using the 'Trust Game' following long-term group psychotherapy. Interest in these relational aspects of PTSD is growing with the introduction of the diagnosis of Complex PTSD (CPTSD) in the new International Classification of Diseases 11 (ICD11) due to be published in 2018; affective and relational functioning are key new components of this new diagnostic category.

The aim of the study was to investigate if and how interpersonal PTSD influences investment behaviour, and therefore possibly trust, compared to individuals without PTSD. We hypothesised that, relative to the healthy controls, the PTSD group would: (a) have lower levels of initial investment (suggestive of trust towards unknown others); (b) behave in a self-protective manner and not increase investments during iterative cooperative interactions; (c) be more sensitive to unfair treatment by others and thus invest lower amounts; and that (d) within the PTSD group, the degree of impairment in trust-related investment behaviour will be influenced by the severity of overall trauma symptoms.

Method

Participants

The sample comprised 54 participants. Participants with PTSD ($n = 22$) were recruited from adult mental health services in South London and Maudsley NHS Foundation Trust and the Helen Bamber Foundation charity; a human rights charity working with survivors of torture and extreme human cruelty. Participants included refugees and asylum seekers who were survivors of state-sponsored torture, human trafficking, slavery, war, and domestic, gender, or sexuality-based violence. Healthy controls ($n = 32$) were recruited via the King's College volunteer database and adverts in the local area. Inclusion criteria for the PTSD group included a primary diagnosis of PTSD given by their treating team as defined by DSM-5 (APA, 2013) with no change in psychotropic medication in the last month. The control group had no current psychiatric diagnosis or history of PTSD. All participants were 18 years+ and required to have a sufficient command of the English language to take part in the study. Criteria for exclusion were: a primary diagnosis of alcohol or substance dependency; organic syndrome or learning disability; or a current risk of suicide/risk to others judged by the clinical team to be sufficient to warrant exclusion.

Outcomes

Post-traumatic Diagnostic Scale (PDS)

PTSD symptoms were confirmed with the PDS (Foa *et al.*, 1997), which was also used to exclude PTSD symptoms in controls. The questionnaire consists of four parts: parts 1 and 2 cover exposure to a range of traumas in accordance with DSM criteria, part 3 covers the severity of symptoms, and part 4 assesses the impact of the trauma on social functioning. It yields a total score (ranging from 0 to 51), with severity ratings: 0 none, 1–10 mild, 11–20 moderate, 21–35 moderate to severe and 36+ severe. The PDS is shown to have a good concordance with full clinical interviews (SCID) to diagnose PTSD.

National Adult Reading Test (NART)

The NART was used as an indicator of general cognitive ability (Crawford *et al.*, 2001) and is widely used in research and is shown to be a good estimate of the intelligence quotient. It was administered to screen for potential differences in IQ between individuals with and without PTSD, to ensure the impairment in trust was not confounded by intelligence.

'Trust Game'

We used a modified version (see King-Casas *et al.*, 2005; Gromann *et al.*, 2013), where participants were told that they would play with two different human partners. Participants were given £10 at the beginning of each trial and could invest any amount between £0 and £10. The pre-programmed partner returned an amount depending on a computer algorithm; the cooperative partner strongly reinforced cooperation, as the chance of a higher repayment increased with every increase in investor trust; and the unfair partner, in turn, reduced repayment with increases in investor trust. In total, all participants played 20 game trials and 20 null trials. Null rounds were designed to ensure participants attended to the game; in these rounds, a red triangle appeared beneath boxes representing different monetary values and the participant was required to move the arrow key to that box. These control rounds were later excluded from the data analysis, after ensuring that participants had correctly understood the task.

Procedure

Participants read the study information sheets and provided written informed consent during an initial session that also included a

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brief interview about demographic information. Participants were assessed individually; they completed the questionnaires, and then the order in which the 'Trust Games' were administered was counterbalanced. Participants were led to believe they were playing against two anonymous persons, in a different room through a computer interface. They were all compensated £25 for their time. The specific focus on trust and PTSD was not explained until the end of the study in order not to bias the sample during the assessment. The study was approved by the Outer West London Research Ethics Committee 10/H0709/8.

Statistical analysis

Analyses were performed in STATA version 14.1 (StataCorp, 2015). Demographics were analysed with $\chi 2$ tests or t tests as appropriate. We analysed the 'Trust Game' data using mixed effects random regression (XTMIXED) to account for repeated measures within a person. The models included the effect of group (PTSD *v.* control), condition (cooperative *v.* unfair), trial number and all two and three-way interaction terms. A dose–response relationship between trauma (PDS scores) and trust in patients was explored using mixed effect random regression.

Results

Sample characteristics

Participants ($n = 54$) were aged 20–63 years (Mean = 34.30, S.D. = 11.47) and ethnicities were white ($n = 25$), black British ($n = 4$), black African ($n = 11$), and other ($n = 14$). They had a NART (Crawford *et al.*, 2001) estimated IQ within the normal (84–123) range. There were no significant differences between the groups in age, ethnicity, or estimated IQ. The PTSD group's mean trauma score (PDS total) was within the 'severe' range (Mean = 39.9, S.D. = 7.8; Range 22–50) and for the controls was 'low' (Mean = 1.9, S.D. = 2.9; Range 0–7).

'Trust Game' performance

The analysis showed a non-significant trend effect for baseline trust ($b = 0.96$, $p = 0.09$; see Fig. 1). The controls made slightly higher investments than individuals with PTSD (6.75, S.D. = 2.41 *v.* 5.78, S.D. = 2.63). The mixed effects random regression analysis showed no significant three-way interaction between group, condition and trial number and no interaction between group and trial number. Interactions between trial number and condition ($b = -0.21$, $p < 0.001$) and group and condition ($b = -0.76$, $p < 0.001$) were significant. Post-hoc analyses by condition showed significantly higher investments during cooperation in controls compared to individuals with PTSD ($b = 1.13$, $p < 0.01$; see Fig. 2). All participants increased their investments over trials ($b = 0.08$, $p < 0.001$). The interaction between group and the trial number was non-significant ($p = 0.81$). The investments did not differ significantly between groups in the unfair condition. A significant effect of trial number showed that all participants decreased investments over trials ($b = -0.13$, $p < 0.001$). The interaction between group and the trial number was non-significant ($p = 0.71$, see Table 1 for mean investments by condition and group). Within patients higher PDS scores were associated with lower trust towards the cooperative ($b = -0.03$, $p < 0.001$) but not towards the unfair other ($p = 0.3$).

Discussion

We examined trust-related behaviour and the ability to develop such trust behaviour in PTSD following interpersonal trauma, using real-time social interactions. The results revealed lower basic trust in those with PTSD compared with controls, but this was not statistically significant. Trial-by-trial interactions showed that both groups were able to anticipate and act upon the behaviour of their partner by adapting their responses accordingly. The PTSD group had a lower propensity to reciprocate their partners' trustworthy behaviour, even in the face of ostensible cooperative behaviour. We considered it probable that interpersonal trauma would skew judgements about others to the negative and this is what was found; higher trauma scores were correlated with lower levels of overall trust-related behaviour. An ability to recognise when to employ trusting behaviour (or not) is positive, but caution developing trust and cooperation may not only influence social relationships but also therapeutic alliances in mental health services.

The smaller initial investments made by the PTSD group may suggest an impaired ability to initiate mutually trusting relationships which fits with the general existing literature that interpersonal traumas are particularly damaging to social relationships (Chouliara *et al.*, 2017). Definitions of trust emphasise that it is an ability (and confidence in the ability) to predict others' behaviour, and it is plausible that this confidence may have been shattered following traumatic events. The modification of responses according to their partners playing style shows an ability to interpret social cues accurately and to anticipate future behaviour, as well as a more general flexibility in behaviour, which has been associated with activation in Theory of Mind related brain areas using the 'Trust Game' paradigm (King-Casas *et al.*, 2008; van den Bos *et al.*, 2011).

Although those with PTSD were able to use feedback to change behaviour, they displayed a lower propensity to trust cooperative others compared to controls. There are several possible explanations. First, although the PTSD group might have a desire to cooperate, they may have been unwilling to reciprocate trust unless they believed that others would not take advantage of their own cooperation; perhaps indicating reduced perspective-taking or adopting a 'better safe than sorry' approach toward others. This fits with the literature that continues to implicate an individual's beliefs as central to the development and maintenance of PTSD (Beck *et al.*, 2014). This is also supported by the dose–response relationship between PDS scores and trust during cooperation. Second, a lower propensity to reciprocate their partners' cooperative behaviour could also be explained by a lack of trust in oneself (rather than the partner). Research indicates that some survivors of interpersonal violence blame themselves for the trauma happening (Deprince *et al.*, 2011), which may place doubt in their mind about their ability in deciding who to trust. Furthermore, they may believe they are not worth achieving the profit and therefore have less interest in cooperating with a partner. One study investigated PTSD symptoms at one and four months post-trauma and reported that the relationships between early re-experiencing and emotional numbing symptoms and later PTSD symptoms were mediated by negative cognitions about the self (Carper *et al.*, 2015). Third, neuroimaging studies with healthy populations have shown activation in brain areas related to reward learning when adapting behaviour during the 'Trust Game' circumstances (King-Casas *et al.*, 2005, 2008; Sanfey, 2007; Phan *et al.*, 2010). Therefore, it is possible that

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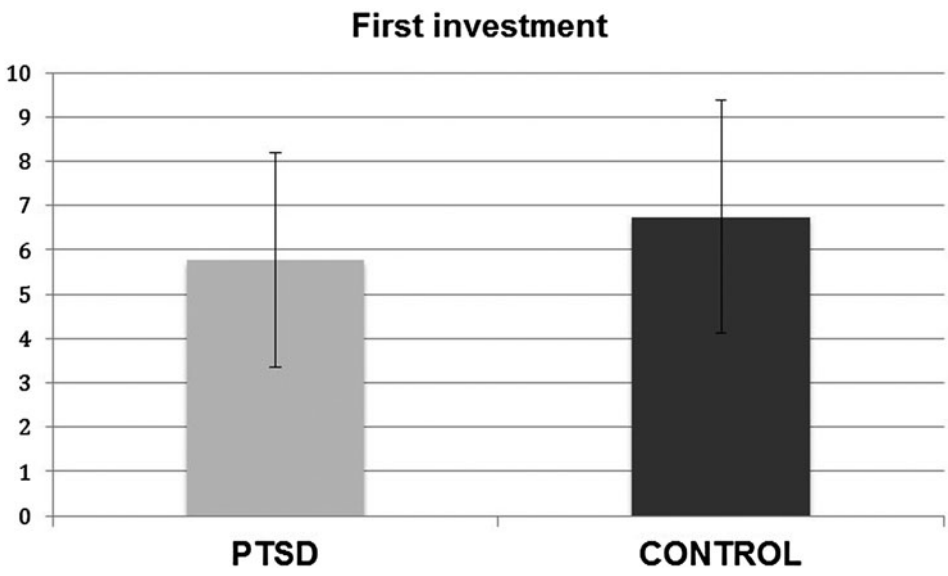

Fig. 1. Non-significant trend effect for baseline trust for both groups.

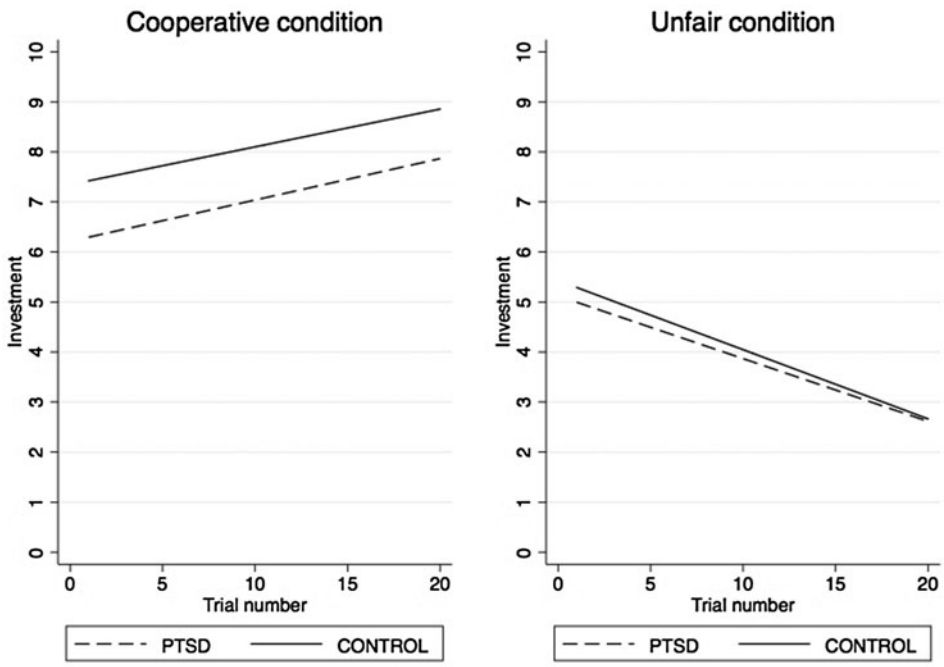

Fig. 2. Investments during cooperative and unfair conditions for both groups.

Table 1. Mean investment by group and condition

Condition	Overall investment £ *M* (S.D.)
Cooperative	
PTSD	7.08 (2.78)
Control	8.14 (2.30)
Unfair	
PTSD	3.81 (2.80)
Control	3.98 (3.37)

the observed finding is due to a reduced experience of social reward following positive social contact, which may limit the reciprocal interactions of PTSD patients with others; future studies could conduct the 'Trust Game' in the MRI scanner, as has been done with those with psychosis, to identify underlying neural processes (Gromann *et al.*, 2013).

During the unfair games, all participants invested less than during cooperative rounds and this is consistent with previous studies (King-Casas *et al.*, 2005; Fett *et al.*, 2012*a*, 2012*b*) that show individuals are sensitive to the violation of the expected reciprocity. Most conceptualisations of trust emphasise that it involves beliefs about others' benevolent motives during social

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interactions that involve a conflict of interests. Balliet and Van Lange (2013) found that the degree of conflict in the dilemma moderated the relationship between trust and cooperation; situations of high conflict make it especially likely that people will condition their cooperation based on beliefs about other's intentions (Holmes, 2004; Simpson, 2007). We anticipated that individuals with PTSD would have reacted more strongly to the unfair partner, but perhaps this condition contained a low conflict of interest and therefore behaviour may have been more influenced by other more self-serving motives, such as impression management. Furthermore, Parks and Hulbert (1995) assessed trust with the public goods game and resource dilemma game, and found that when fear was present (i.e. greater costs associated with a partner taking advantage of one's own cooperation), those with initial high trust cooperated more frequently than those with initial low trust; however when fear was absent (i.e. when a partner's non-cooperative behaviour caused less harm to one's own outcomes), both groups performed the same. Given the importance placed on a sense of current threat in cognitive theoretical models of PTSD, these results imply that the findings may be influenced by whether the particular social dilemma contains an element of fear and this could be explored in future work.

We also found that those experiencing the most severe trauma symptoms made the lowest investments indicating a greater impairment in trust, as measured by the average investments during the cooperative trials. Having more severe symptoms could indicate a more recent exposure to trauma or prolonged exposure over a longer period of time; perhaps these individuals are more likely to re-experience the traumas more frequently than those with milder symptoms and thus the inability to develop trust is maintained. It will be important in future work to clarify the role of trust and reciprocity in the onset, maintenance, and complexity of PTSD.

A key strength of the study is that it illustrates the potential methodological advancement in the use of a behavioural economic paradigm to study relational trust in PTSD. Nevertheless, the study has some limitations worth noting in order to enhance the impact of future research. First, it is not possible to determine cause and effect due to the cross-sectional design; the inclusion of a comparison accidental trauma sample and a larger sample size would be needed to replicate these findings more robustly. Second, one advantage of probabilistic tasks, like the 'Trust Game', is that analysis of learning in terms of prediction errors is possible, this would involve elaborating a model of the prior, the prediction error mechanism, and how action selection (investment amounts) is dependent on the prediction error mechanism. This aspect of the data has not been explored in this study, primarily because there are too few data points to successfully fit this type of model. However, a future study with larger number of trials could usefully examine these interesting parameters. Third, it could be argued that the findings related to a lower propensity to reciprocate partners' trustworthy behaviour are a result of high levels of anxiety or depression; therefore assessment of comorbid disorders will be important. It may also be beneficial to corroborate the 'Trust Game' data with other behavioural measures of trust alongside a self-report questionnaire of relational trust. Fourth, it is possible that the 'Trust Game' might depart too much from the type of trust decisions that survivors of interpersonal trauma make about others and which often include vulnerability or the threat of harm from others. Implementing a dynamic manipulation of reciprocity from the partner could provide a context within which to examine how the participants respond to unexpected changes in social behaviour. Participants were primed with the idea that they were playing with another person, and this concept is widely recognised in psychology to influence people's decision-making even if it happens subconsciously (Bargh, 2006). However, it is possible that participants did not believe they were playing against a human partner, and future investigations should provide an appropriate level of detail about the partner's identity. For example, trustworthiness is dependent on a person's appearance, ethnic origin and the situation in which trust is exerted. This raises the broader question of the extent to which the 'Trust Game' reflects actual experiences of trust in day-to-day life and future studies need to address how behaviour in the Trust Game is associated with social behaviour in real life, for example with the experience sampling measures. Finally, we are curious about the association between severity of trauma symptoms, perceived extent of betrayal that occurred during the trauma (Martin *et al.*, 2013) and the experience of social defeat that may increase the risk of developing paranoid appraisals (Valmaggia *et al.*, 2015). Investigating paranoia in response to interpersonal trauma is a novel focus for research, looking at the extent to which fears are generalised and exaggerated beyond the trauma events. The hypothesis here would be that clinical paranoia may emerge in those with PTSD as a result of a loss of trust in others, so that loss of trust may contribute both to the PTSD and the paranoia. One plausible way forward is to assess the degree to which these same psychological processes may occur in PTSD, psychosis and non-clinical samples, as consistent with assumptions of a continuum (Bebbington *et al.*, 2013).

In conclusion, these findings provide preliminary support to our main hypothesis that diminished trust may be an important process in relational difficulties following interpersonal PTSD. We found trust in PTSD was only significantly lower when playing cooperative others, and that higher PDS scores were associated with lower trust but again only during cooperative tasks. Therefore, the factors that contribute to the development and maintenance of reciprocating cooperative behaviour could provide a focus for future cognitive and pharmacological interventions.

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Conflict of interest. None.

Ethical standards. The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Helsinki Declaration of 1975, as revised in 2008. The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional guides on the care and use of laboratory animals.

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